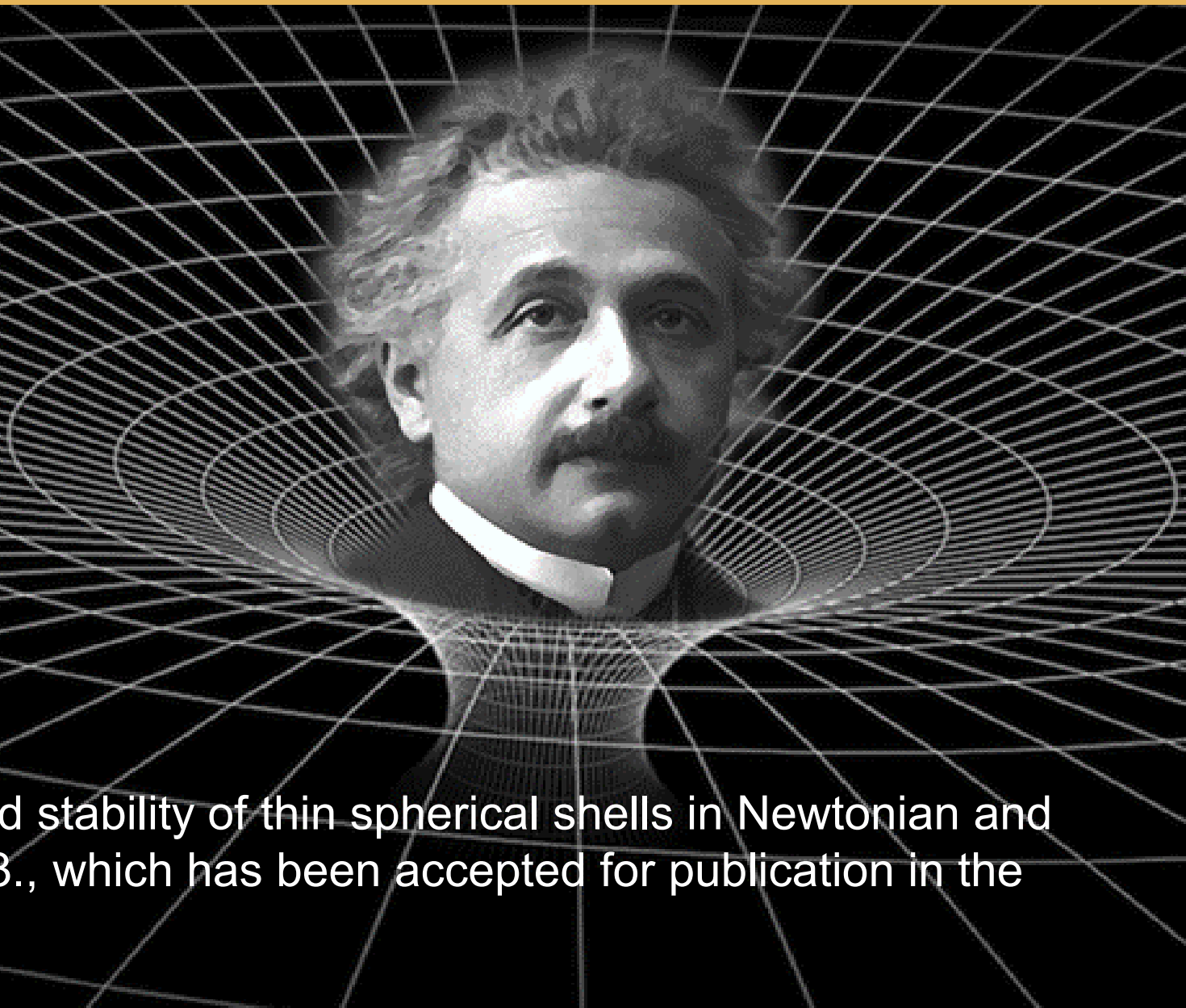
UNIVERSITY of GUELPH

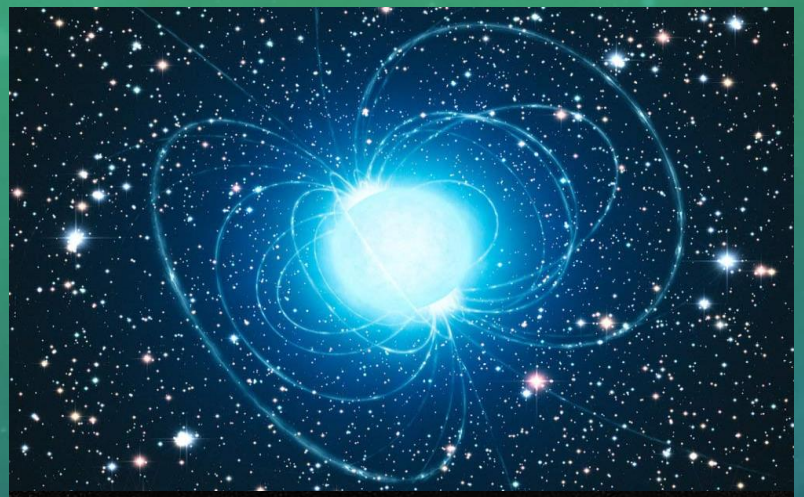
Thin Shells as Neutron Stars: A Simple Stellar Model with Great Insight

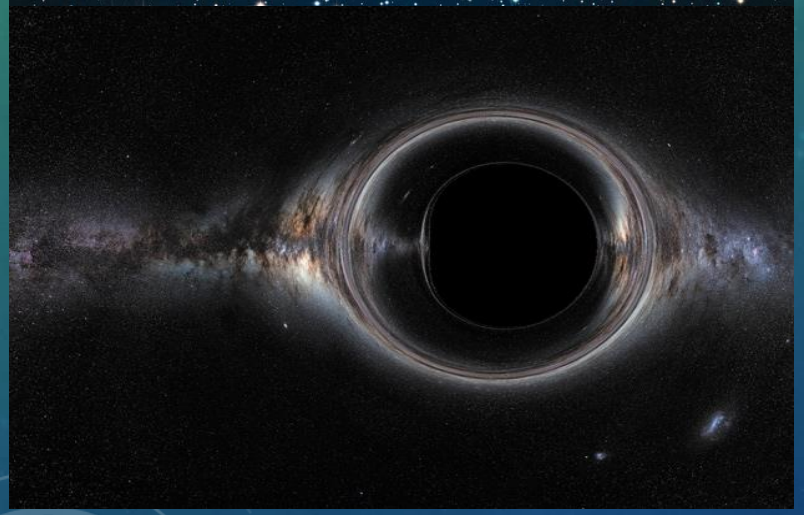


Phil LeMaitre, Eric Poisson

Based on our paper: “Equilibrium and stability of thin spherical shells in Newtonian and relativistic gravity”. arXiv:1909.06253., which has been accepted for publication in the American Journal of Physics

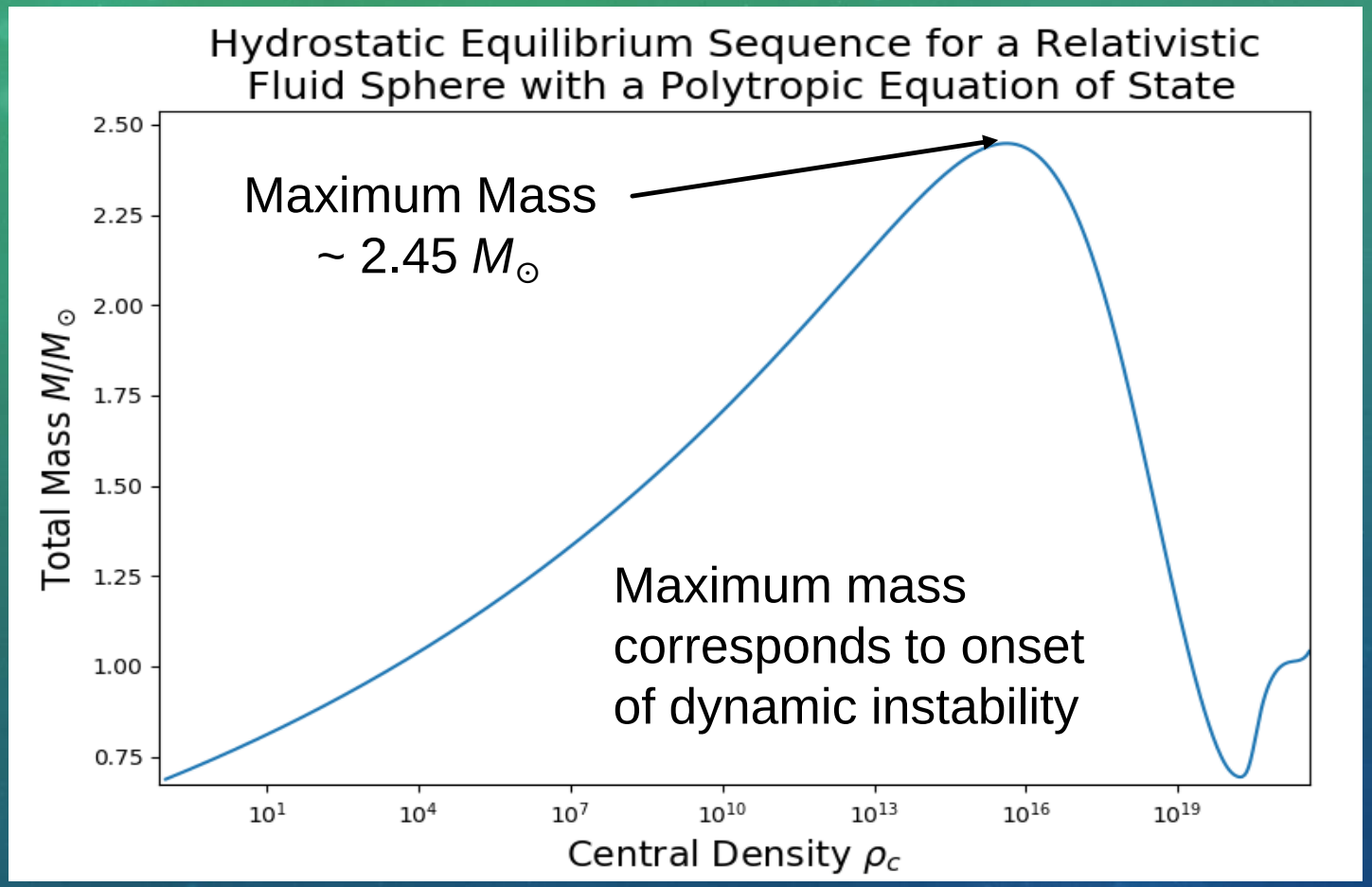
Compact Objects



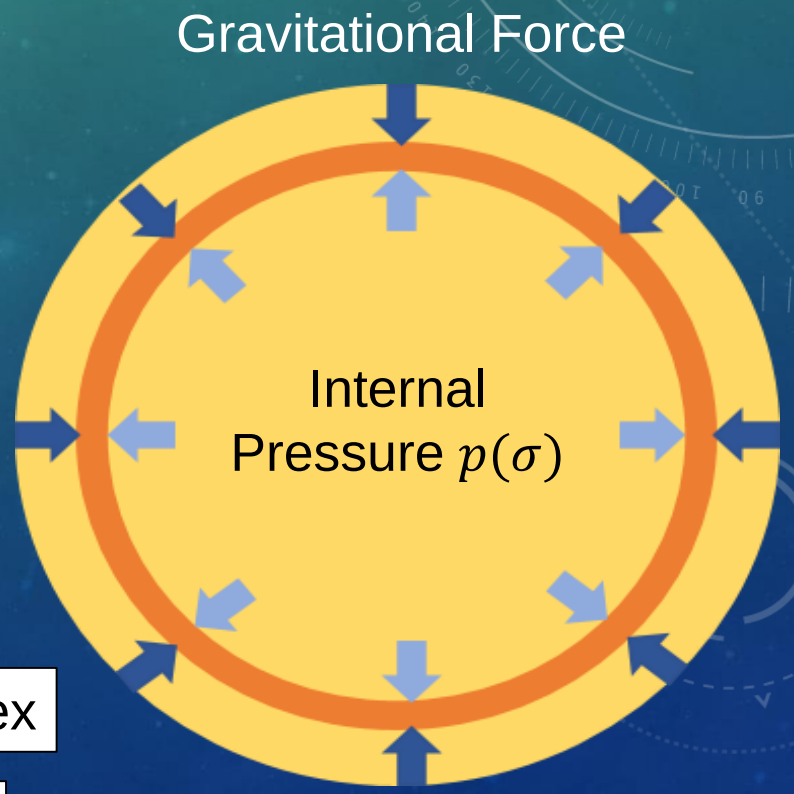


Compact Object	Mass	Radius	Location
Neutron Star	$1 - 2.5\ M_\odot$	~ 10 km	Binary systems and interstellar nebulae
Stellar Mass Black Hole	$5 - 100\ M_\odot$	$15 - 300$ km	Binary systems and interstellar nebulae
Intermediate Mass Black Hole	$\sim 10^3\ M_\odot$	$\sim 10^3$ km	Interstellar space and globular clusters
Supermassive Black Hole	$10^5 - 10^{10}\ M_\odot$	$10^{-3} - 400$ AU	Galactic Centres

Fluid Sphere Neutron Star Model



- Hydrostatic Equilibrium ($\ddot{R} = \dot{R} = 0$): Fluid rest frame, meaning internal pressure balances external force of gravity



Polytropic Equation of State:

$$p(\sigma) = K\sigma^{\gamma}$$

Constant of Proportionality

Polytropic Index

Mass Density

Why Thin Shells?

- Governing equations and equilibrium configurations are relatively simple
  - Fluid sphere equilibrium sequences require numerical integration of 2 parametrized equations of motion and several series expansions about the origin
- Maximum mass link to onset of stellar instability is demonstrated easily using only undergraduate algebra and calculus
  - Fluid sphere model requires relativistic fluid perturbation theory
- More accessible to introductory relativity students
  - Almost all results can be expressed analytically

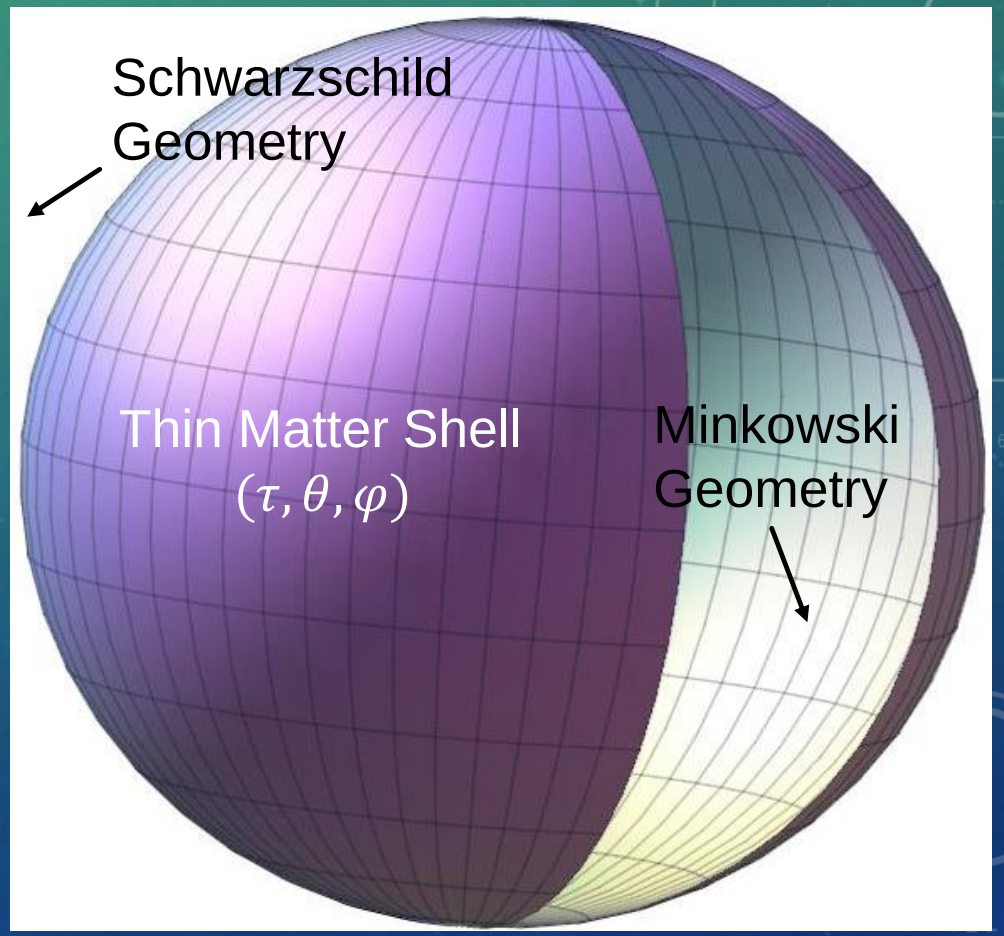
Thin-Shells

- Interior Minkowski Geometry: Flat spacetime
- Exterior Schwarzschild Geometry: Vacuum spacetime outside of a spherically-symmetric, uncharged, and static mass
- Perfect-fluid (i.e. no heat conduction, shear stress, or viscosity) matter content of surface smoothly joins both geometries
- We use the reference frame co-moving with the shell surface
- The shell is macroscopically described by it's compactness:

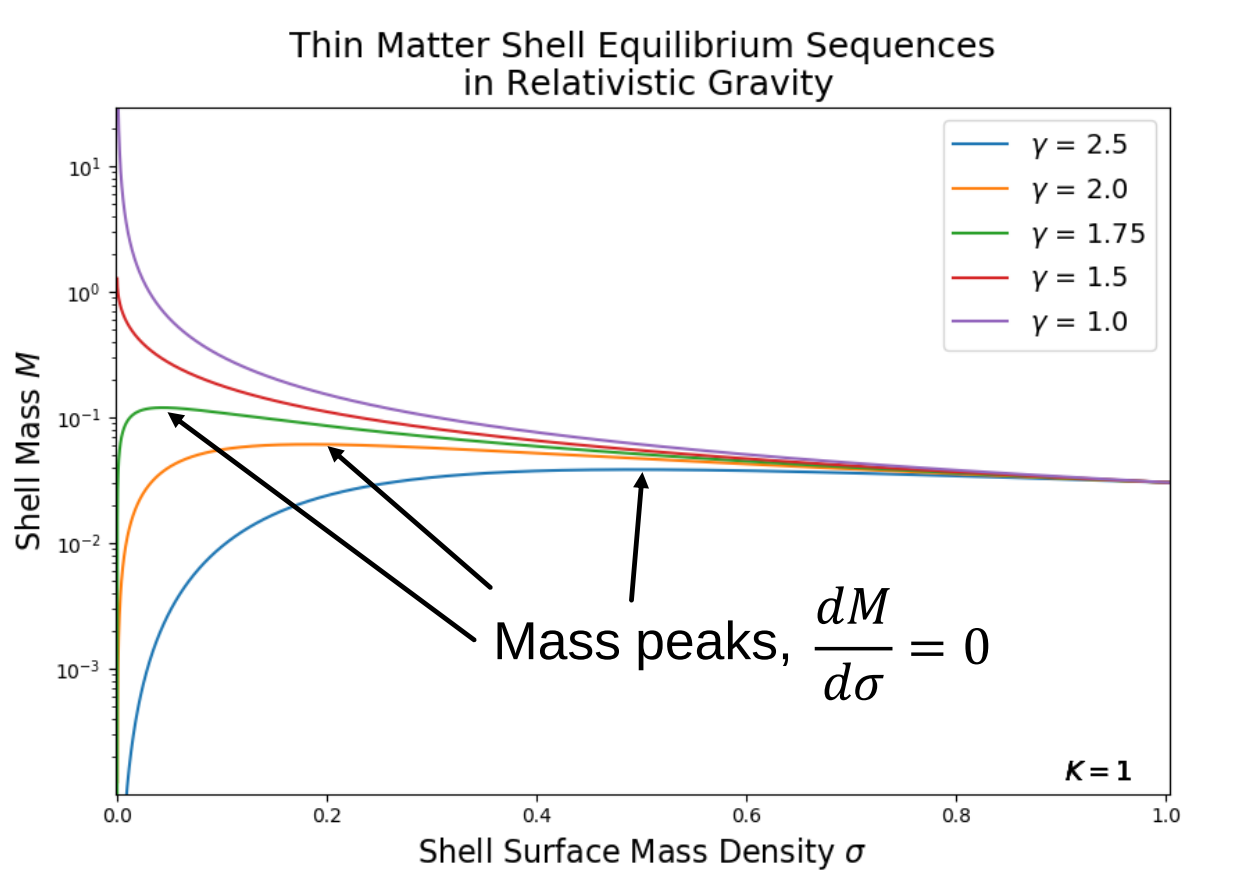
$$C = \frac{m}{R}$$

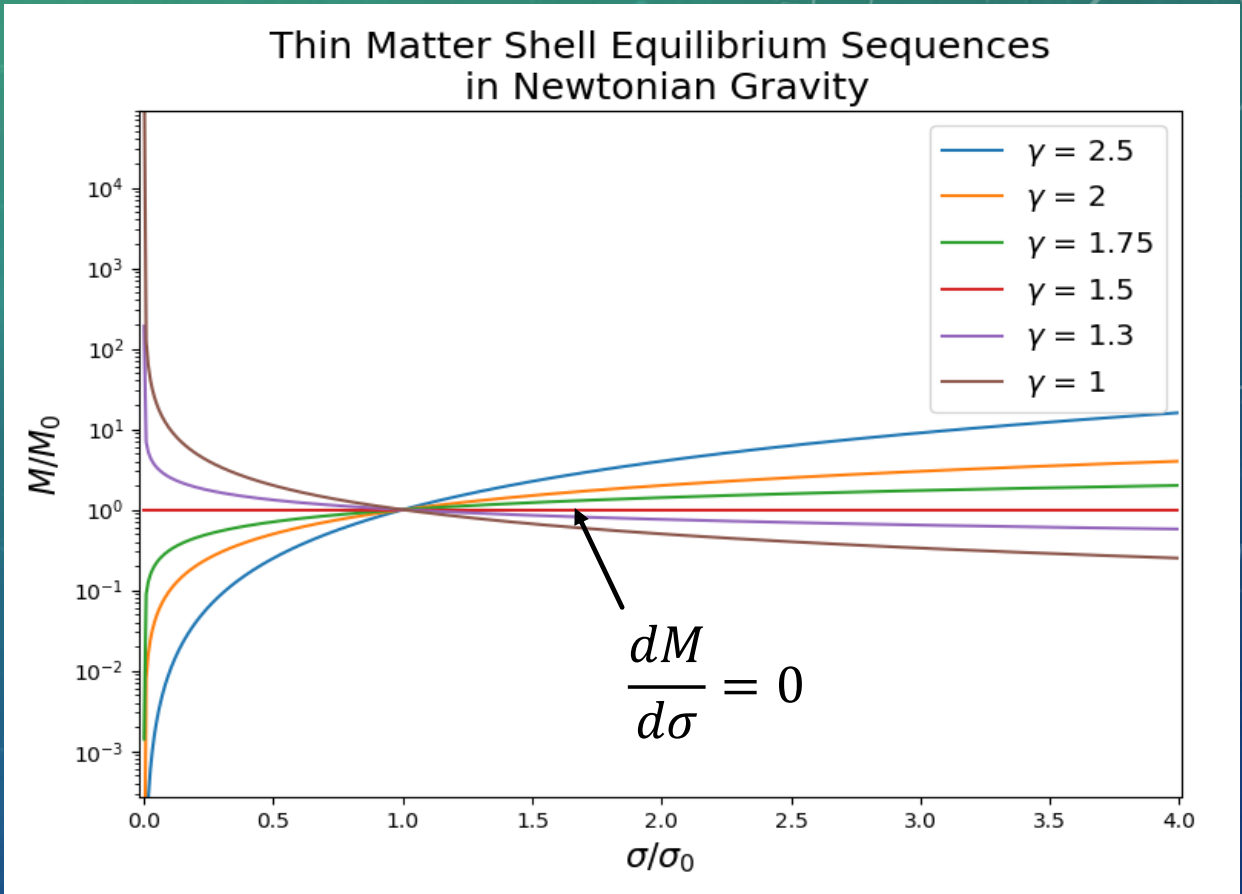
Material Mass

Shell Radius



Hydrostatic Equilibrium for the Relativistic and Newtonian Thin Matter Shells





- $\frac{dM}{d\sigma} = 0$ corresponds to a variable turning point:

$$\gamma = \frac{3C^2 - 8C + 6}{(C-1)(3C-4)}$$

- $\frac{dM}{d\sigma} = 0$ corresponds to a constant turning point:

$$\gamma = \frac{3}{2}$$

Stability Conditions

For small radial perturbations $|\delta R| \ll 1$, the solution has the form of a harmonic oscillator:

$$\boxed{\delta R = Ae^{\pm i\omega\tau}},$$

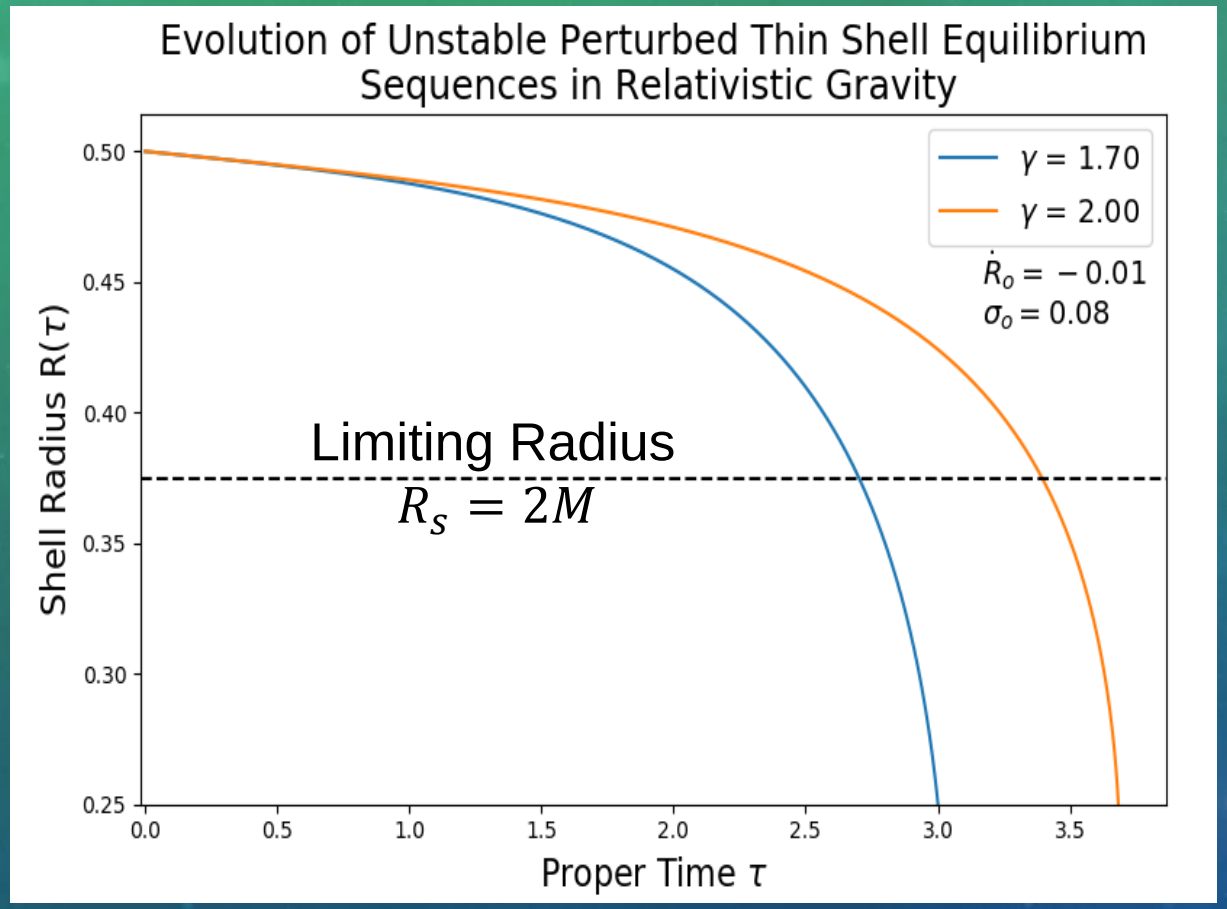
where A is a constant determined by initial conditions ($\tau = t$ in Newtonian regime).

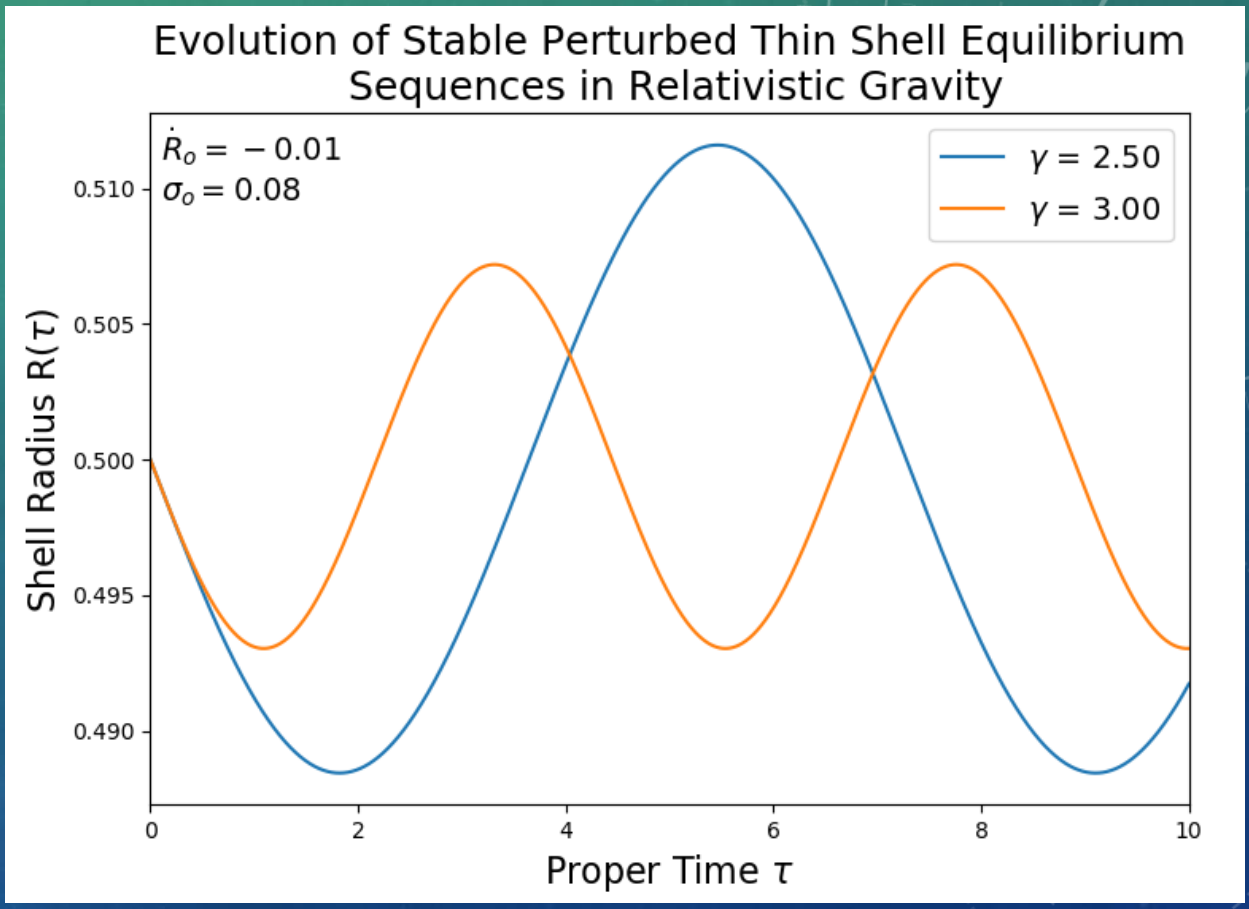
	Newtonian Shell	Relativistic Shell
Harmonic Oscillator Frequency	$\omega = \sqrt{\frac{GM_o}{R_o^3}\left(\gamma - \frac{3}{2}\right)}$	$\omega = \sqrt{\frac{C^3}{4m_o^2(1-C)^2}\left((\gamma-3)(C-1)\left(C-\frac{4}{3}\right)-C\right)}$

Newtonian Gravity	Relativistic Gravity	ω	δR	Shell Behaviour
$\gamma > \frac{3}{2}$	$\gamma > \frac{3C^2-8C+6}{(C-1)(3C-4)}$	Real	$Ae^{\pm i\omega\tau}$	Oscillating
$\gamma < \frac{3}{2}$	$\gamma < \frac{3C^2-8C+6}{(C-1)(3C-4)}$	Complex	$Ae^{\pm\omega\tau}$	Expansion/Contraction
$\gamma = \frac{3}{2}$	$\gamma = \frac{3C^2-8C+6}{(C-1)(3C-4)}$	0	A	Constant

- $\gamma_{Rel} > \gamma_{Newt}$ to maintain shell stability since relativistic gravity is stronger than Newtonian gravity
- For $C \ll 1 \rightarrow \gamma \approx \frac{3}{2}$; Newtonian gravity is recovered!

Thin-Shell Perturbations From Equilibrium

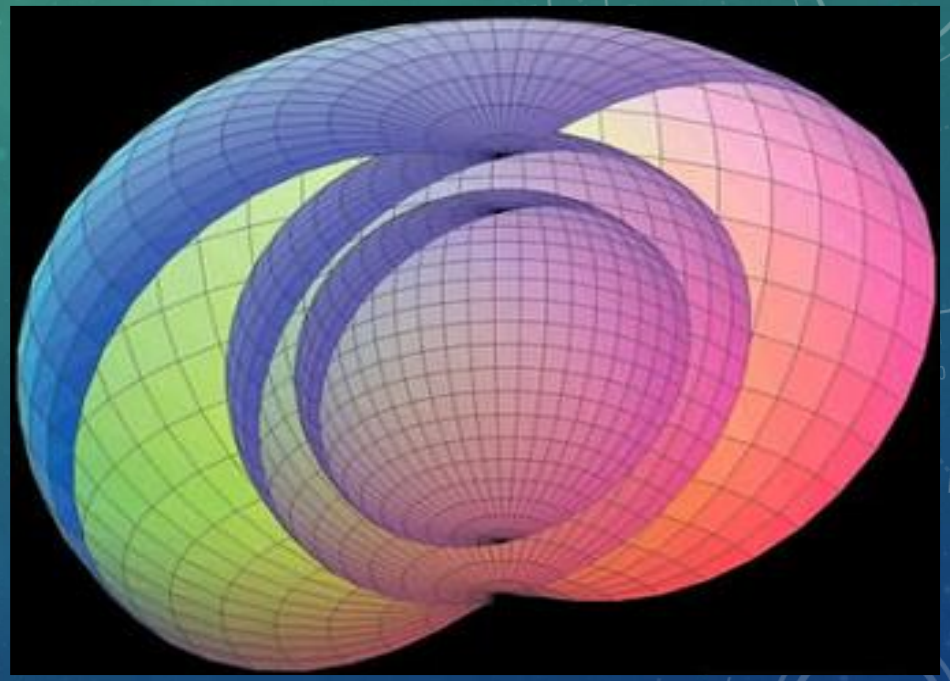




- For an unstable γ, thin-shell transitions to a black hole once the limiting radius is reached
- For a stable γ, thin-shell oscillates until perturbed again

Further Analysis

- Apply the methodology to a non-perfect fluid thin-shell
- Explore the behaviour of concentric thin-shells and thin-shell crossings
- Apply the methodology to a thin-shell connecting different geometries
- Investigate different equations of state
- Extend the research to higher and lower spacetime dimensions

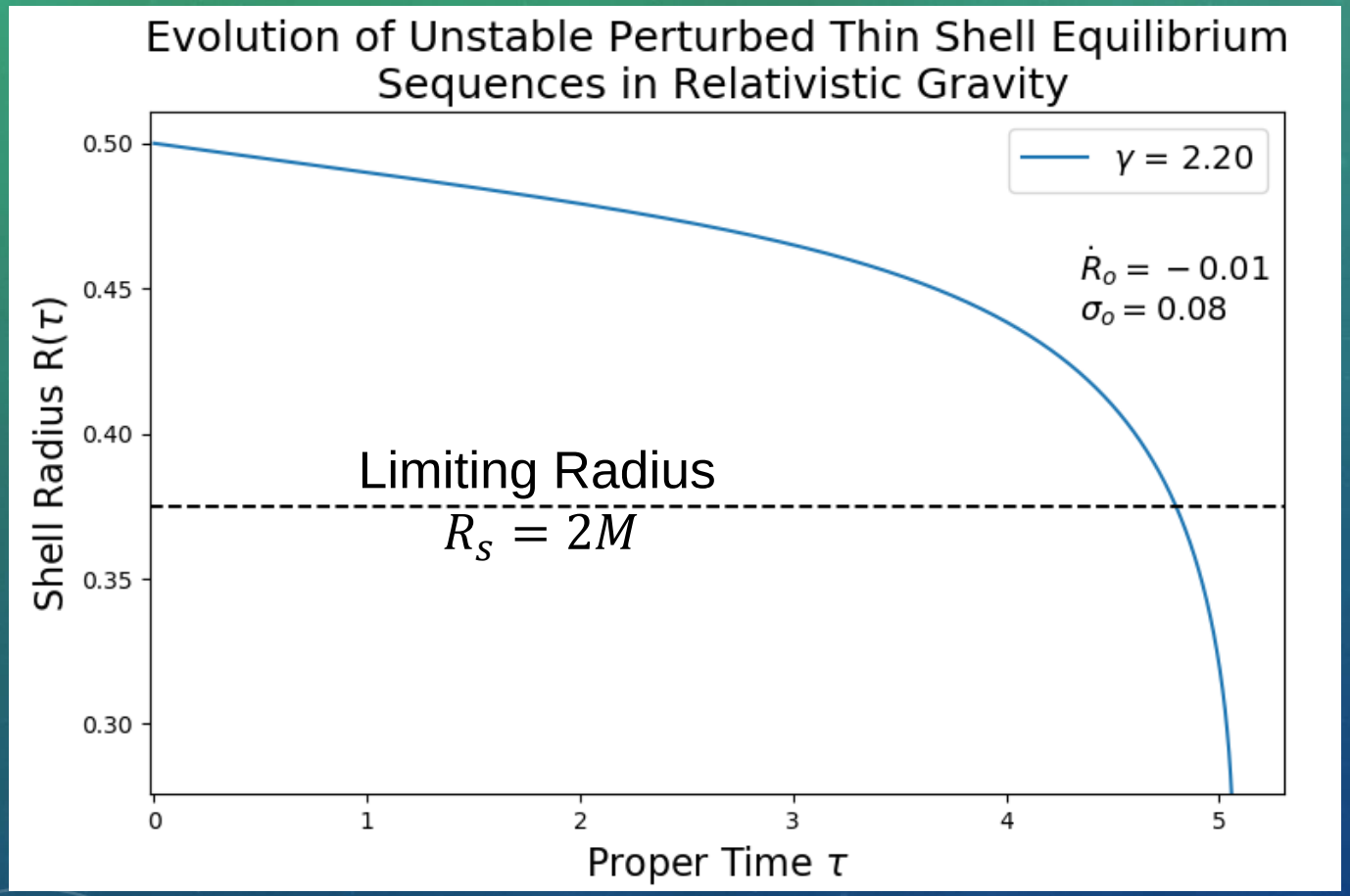


Thank you for Listening!
Merci de Votre Attention!

Hydrostatic Equilibrium for the Relativistic and Newtonian Thin Matter Shells

- In relativistic gravity $\frac{dR}{d\sigma} = 0$ corresponds to a variable turning point: $\boxed{\gamma = \frac{2-C}{1-C}}$
  - Since we assume that R should decrease for increasing σ, an upper bound for γ is this turning point
- In Newtonian gravity $\frac{dR}{d\sigma} = 0$ corresponds to a constant turning point: $\boxed{\gamma = 2}$

Thin-Shell Perturbations From Equilibrium



- The ‘constant’ solution curve is unstable for larger perturbations